Milestone 1 Report – SmartLink Software Design Project: SmartLink – Professional Networking Platform

1. Team Members

Name	ID
Omar Khaled Hussein	22010962
Ayman Ibrahim Mohamed Kotb	22010656
Ali El-Deen Maher Abd El-Moniem	22010934
Ali Mostafa Mohamed	22010939
Yaseen Asaad Ahmed Farid	22011349
Farouk Ashraf Farouk	22011012

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2. Milestone Objective

The primary goal of Milestone 1 is to establish the technical and functional foundation of the SmartLink platform. This includes setting up the development environment, database schema, and implementing core authentication along with basic social interaction features to prepare the system for future expansion.

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3. Scope & Implemented Features

3.1 Authentication & User Management

- User Sign Up (Email & Password)

- User Sign Up using Gmail (Google OAuth)
- User Login (Email/Gmail)
- Admin Login with restricted privileges
- Basic Profile Management

3.2 Feed & Social Interaction

- Professional Feed Display
- Create Post
- Edit Post
- Like & Comment on Posts

3.3 Company Profile Management

- Create Company Profile
- View Company Profile & About Section
- Update Company Information
- Manage Multiple Company Locations
- Track Company Followers

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4. Functional Requirements

4.1 User Registration (Basic)

Users register with:

- Full Name
- Email (unique)
- Password & Confirm Password
- Birth Date
- Phone Number (unique)
- Gender

Validation Rules:

- Valid email format
- Strong password enforcement

- Age must be between 13 and 120
- Real-time error feedback

Output:

- Successful registration confirmation
- Error messages if validation fails

Priority: High

4.2 User Registration (Gmail)

Users can register using Google authentication for faster access and reduced manual input.

Priority: High

4.3 User Login

- Secure login using Email/Password or Gmail
- Proper error handling for incorrect credentials

Priority: High

4.4 Admin Login

- Separate admin authentication interface
- Elevated access permissions

Priority: High

4.5 View Company Profile

Users can view company information:

Inputs:

- Company ID
- User ID

Displayed Information:

- Company (name, industry, website)
- Description
- Logo & Cover image
- Number of followers
- locations
- Follow status

Output:

- Complete company profile view
- "Follow" or "Following" button state
- Company about section
- Company posts

4.6 Update Company Profile

Company owners can update their profile information:

Editable Fields:

- Company Name
- Description
- Website
- Industry
- Founded Year
- Logo URL
- Cover Image URL
- Locations
- Only the company owner can update

Output:

- Updated company profile data
- Updated locations reflected immediately
- Success/error messages

4.7 Follow Company

Users can follow companies to stay updated:

Inputs:

- Company ID
- User ID (follower)
- **Validation Rules:**
- Follow relation between user and company added

- number of followers of company increased by 1

Output:

- "Successfully followed company" message
- Updated follower count displayed
- Follow button changes to "Following"

4.8 Unfollow Company

Users can unfollow companies:

Inputs:

- Company ID
- User ID (follower)

Validation Rules:

- Follow relation between user and company deleted
- number of followers of company decreased by 1

Output:

- "Successfully unfollowed company" message
- Updated follower count displayed
- "Following" button changes to "Follow"

4.9 Company Locations

Companies can manage multiple office locations:

Features:

- Add multiple locations (City & Country)
- Remove locations
- Update all locations at once

Validation Rules:

- Both city and country are required
- Empty or whitespace-only values are ignored

- Duplicate locations are prevented

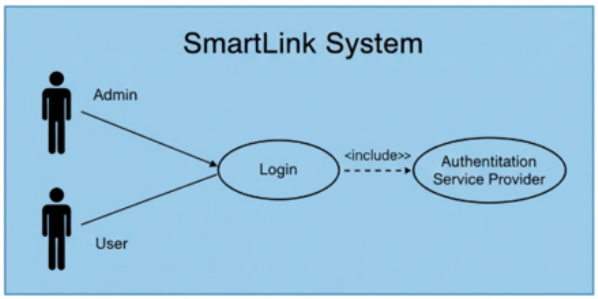
Output:

- Updated list of company locations
- Location data synchronized across system

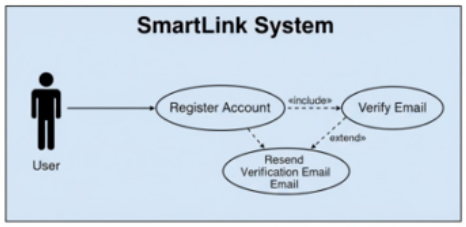
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5. Use Case and Sequence/Activity diagram Summary

Login use case

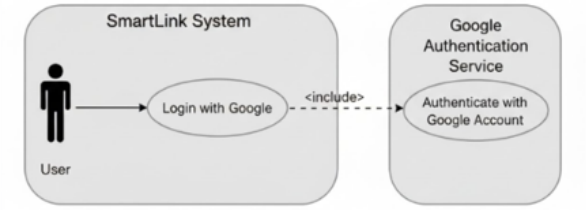


Register Use Case Diagram

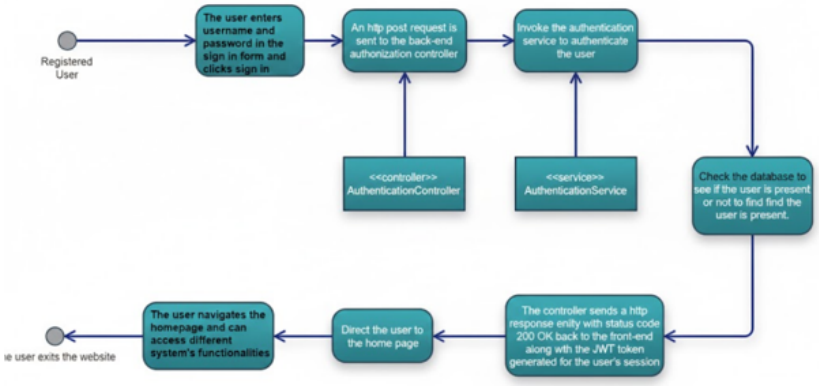


Google Authentication use case

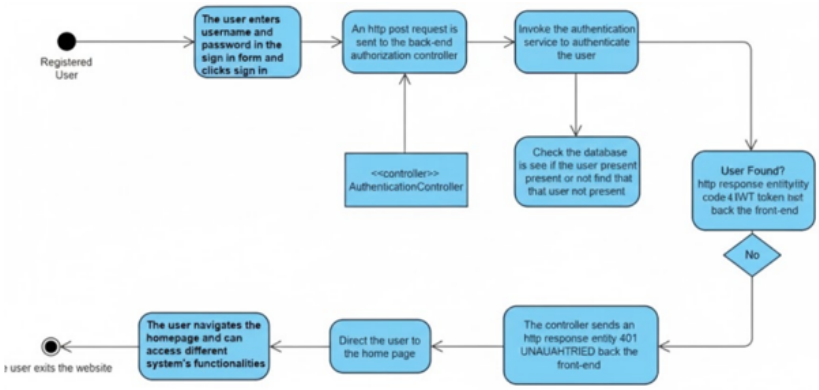
Google Authentication



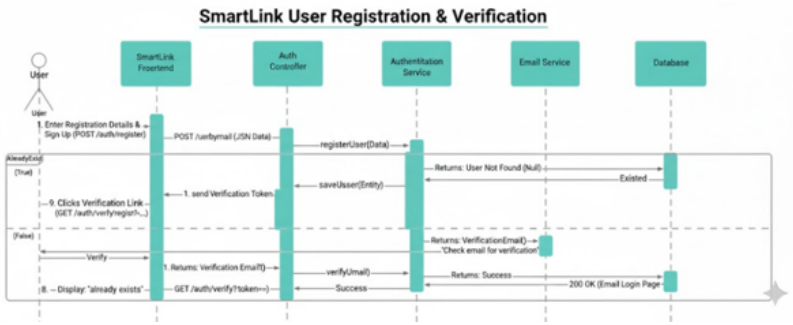
registered user login



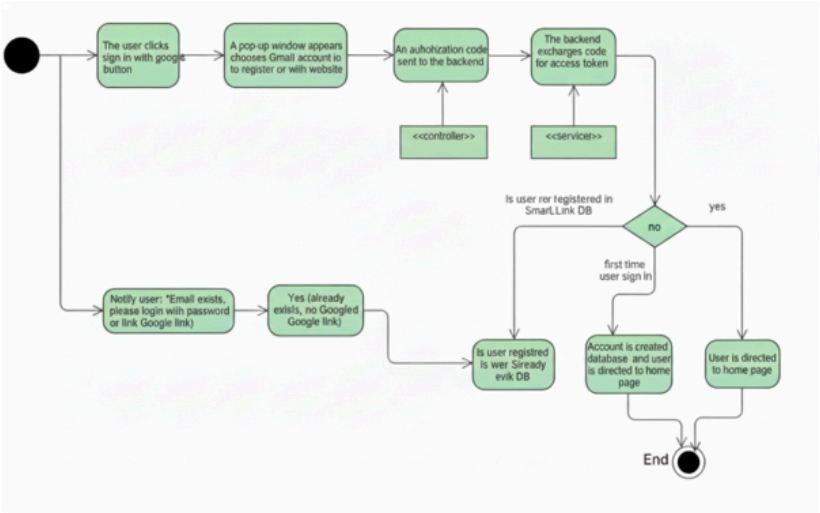
unregistered user login



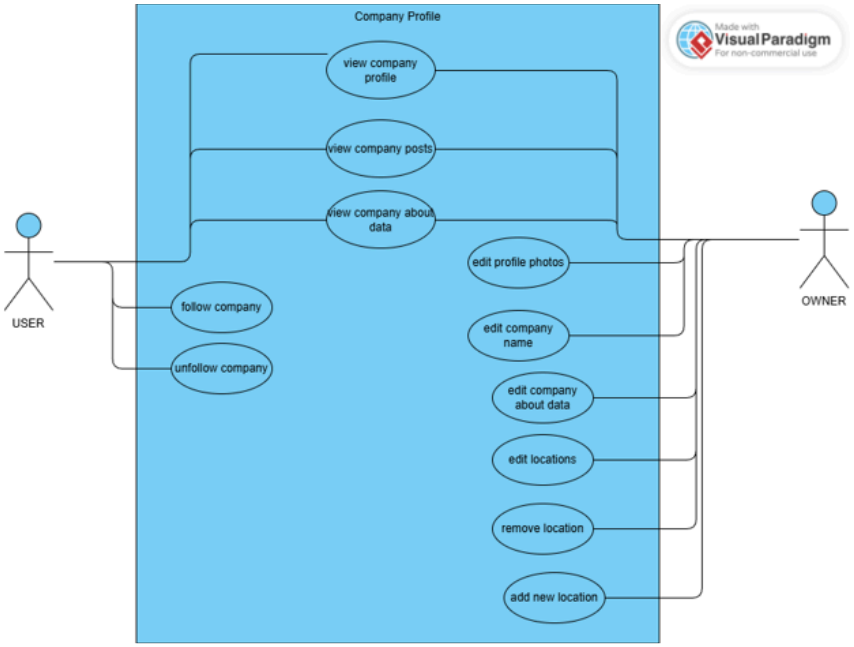
Register Sequence Diagram



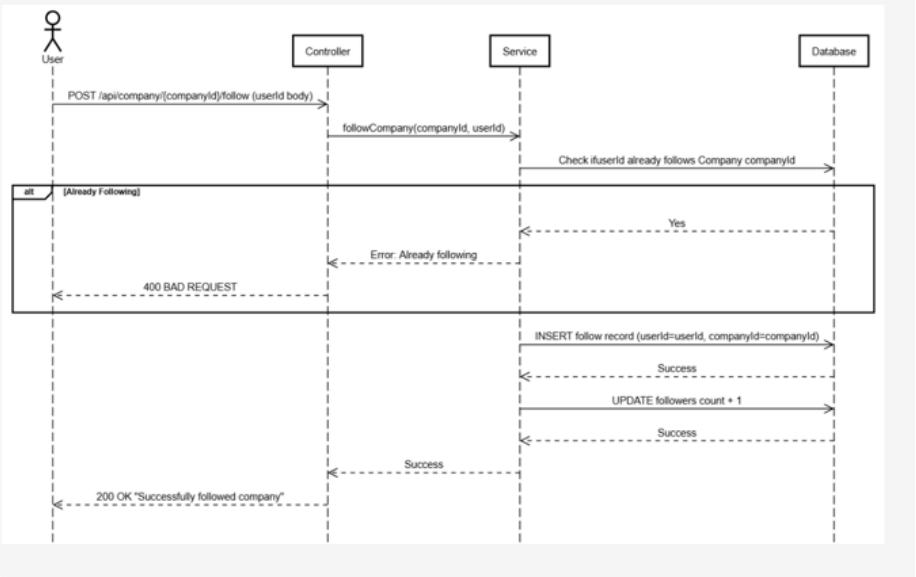
Google Authentication activity diagram



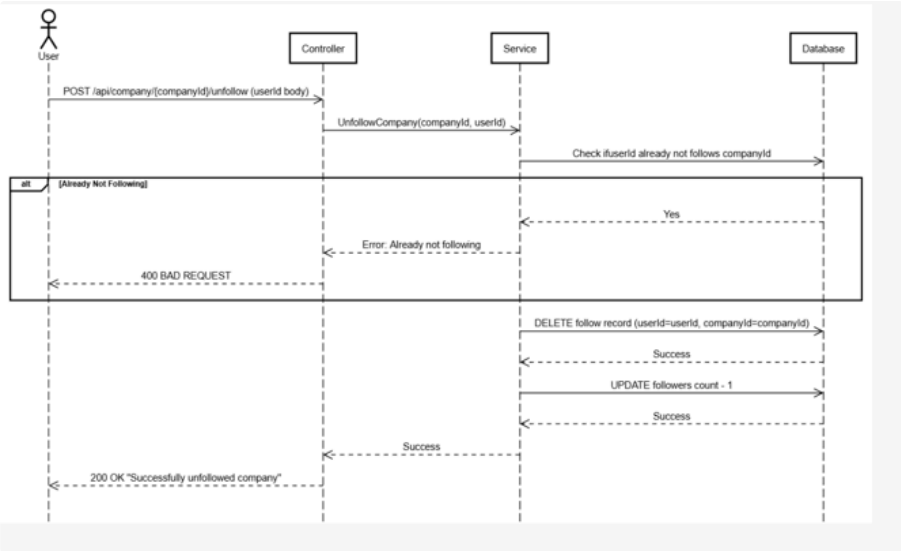
Company Profile use case Diagram



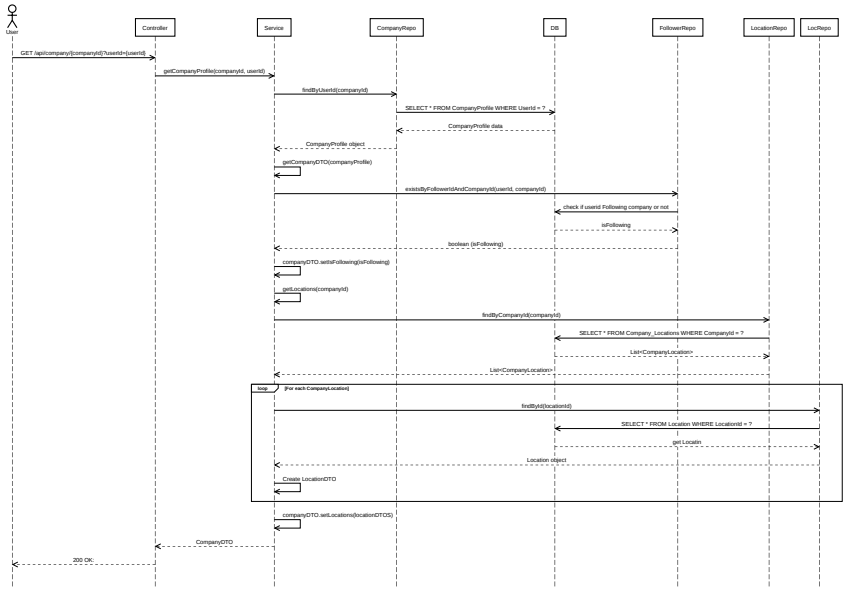
Follow company Sequence diagram



unFollow company Sequence diagram



Get company profile data



6. System Architecture

Initial Design

- Monolithic structure
- Direct UI-to-database communication

Improved Design

- Layered Architecture:
  - Presentation Layer: React
  - Business Layer: Spring Boot Services
  - Data Layer: Repositories & MySQL

Benefits

- Better security
- Clear separation of concerns
- Improved scalability
- Easier maintenance

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7. Workflow & Project Management

Development followed Agile methodology using Jira for task management.

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8. Testing Strategy

Unit Testing

- Backend coverage: more than 70%
- Tools used: JUnit & Mockito
- TDD approach applied

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9. Version Control & Collaboration

- GitHub used for source control [omar2004khaled/SmartLink: Social app with authentication and user management]

- Each member created and reviewed Pull Requests
- All PRs linked to Jira tasks for traceability Smart-Link - Backlog - Jira

Prepared by: SmartLink Development Team